# Thinned Poisson (Experimental)

## Parametrisation

The Poisson distribution is

$$\text{Prob}(y) = \frac{\lambda^y}{y!}\exp(-\lambda)$$

for responses $y = 0, 1, 2, \ldots$, where $\lambda$ is the expected value.

The thinned Poisson allow the observations to have a known or unknown thinning: `event`$= 1$ its observed as is. With `event`$= 0$ (or $\neq 1$) its thinned, so the likelihood is Poisson with mean $p(\cdot)\lambda$ where

$$\text{logit}(p(\cdot)) = \text{offset} + \sum_{i=1} \beta_i x_i$$

## Link-function

The mean $\lambda$ is linked to the linear predictor by

$$\lambda(\eta) = E\exp(\eta)$$

where $E > 0$ is a known constant (or $\log(E)$ is the offset of $\eta$).

## Hyperparameters

$\beta_1, \beta_2, \ldots$ if in use. Maximum 10.

## Specification

- `family="tpoisson"`
- Data are given as an `inla.mdata`-object, with format

  $$\text{inla.mdata}(y, E, \text{event}, \text{offset}, x_1, x_2, \ldots)$$

  where maximum 10 covariates can be given. Each argument is a vector. Note that the four first columns are required, and the covariates can be omitted if there are none.

### Hyperparameter spesification and default values

**doc** `Thinned Poisson`

**hyper**

**theta1**

**hyperid** `66721`
**name** `beta1`
**short.name** `beta1`
**output.name** `beta1 tpoisson observations`
**output.name.intern** `beta1 tpoisson observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`

**from.theta** `function(x) x`

**theta2**

**hyperid** `66722`
**name** `beta2`
**short.name** `beta2`
**output.name** `beta2 tpoisson observations`
**output.name.intern** `beta2 tpoisson observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta3**

**hyperid** `66723`
**name** `beta3`
**short.name** `beta3`
**output.name** `beta3 tpoisson observations`
**output.name.intern** `beta3 tpoisson observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta4**

**hyperid** `66724`
**name** `beta4`
**short.name** `beta4`
**output.name** `beta4 tpoisson observations`
**output.name.intern** `beta4 tpoisson observations`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta5**

**hyperid** `66725`
**name** `beta5`
**short.name** `beta5`
**output.name** `beta5 tpoisson observations`
**output.name.intern** `beta5 tpoisson observations`
**initial** `0`

**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta6**

**hyperid** 66726
**name** beta6
**short.name** beta6
**output.name** beta6 tpoisson observations
**output.name.intern** beta6 tpoisson observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta7**

**hyperid** 66727
**name** beta7
**short.name** beta7
**output.name** beta7 tpoisson observations
**output.name.intern** beta7 tpoisson observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta8**

**hyperid** 66728
**name** beta8
**short.name** beta8
**output.name** beta8 tpoisson observations
**output.name.intern** beta8 tpoisson observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta9**

**hyperid** 66729
**name** beta9

**short.name** beta9
**output.name** beta9 tpoisson observations
**output.name.intern** beta9 tpoisson observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta10**

**hyperid** 66730
**name** beta10
**short.name** beta10
**output.name** beta10 tpoisson observations
**output.name.intern** beta10 tpoisson observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**survival** FALSE

**discrete** TRUE

**link** default log

**pdf** tpoisson

## Example

In the following example we estimate the parameters in a simulated example with Poisson responses.

```
n <- 3000
x <- rnorm(n)
eta <- 1 + 0.2 * x
event <- rep(1, n)
E <- runif(n)

offset <- rnorm(n, sd = 0.3)
xx <- rnorm(n)
xxx <- rnorm(n)
eta.c <- offset + 0.3 * xx + 0.4 * xxx

y <- numeric(n)
prob <- 1/(1+exp(-eta.c))
event <- sample(c(1, 0), n, prob = c(0.6, 0.4), replace = TRUE)
prob[which(event == 1)] <- 1
y <- rpois(n, prob * E * exp(eta))
```

```
Y <- inla.mdata(y, E, event, offset, xx, xxx)
r <- inla(Y ~ 1 + x,
          data = list(Y = Y, x = x),
          family = "tpoisson",
          control.family = list(hyper = list(beta1 = list(param = c(0, 1)),
                                             beta2 = list(param = c(0, 2)))),
          verbose = TRUE,
          debug = TRUE)
summary(r)
```

### Notes